



The Secret World of Scams

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In a world full of scams, how confident are you in identifying false emails, URLs and files? By breaking down the tactics scammers use, I want to help others spot the signs and stay safe online. It's practical, eye-opening, and more relevant than ever.

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INTRODUCTION

Depending on who you ask, and when you ask, almost everyone will have a different answer to the question, "What is a scam?"

When we think of scams, one of the first thought that usually pops into mind is the infamous Nigerian Prince Scam, offering you millions and millions of dollars for the low price of your credit card information. The rapid change in technology has made these obvious far and in between with more sophisticate and deadly scams taking its place.

EVOLUTION OF SCAMS – (800 WORDS)

Scams have taken many different forms throughout human history, but one thing has remained the same: someone has taken advantage of someone else for their own gain.

The Past of Scams

Looking back to history, the earliest recorded scam occurred in 300 BC. A Greek sea merchant, Hegestratos, took out an insurance policy on his ship and cargo. The policy allowed merchants to borrow money as long as they agreed to pay it back with interest once the cargo was delivered. Hegestratos intentionally sank his empty ship, so he did not repay the loaned money. However, he was caught attempting to escape, after sinking his ship. Scams have always, and will always exist, so long as there is money to be made.

Before internet scams became widespread, they occurred in the physical, through confidence tricks. Confidence tricks is a deceptive scheme where someone gains the victims trust before defrauding them. They exploit the victim's naivety, greed and compassion. A famous example is the Ponzi scheme, named after Charles **Ponzi**.

Scam Study: The Ponzi Scheme

Charles Ponzi was a con artist who was one of the first financial criminal of modern times. He secured \$15 million in a few months from Boston locals by convincing them to invest in his fake invest scheme. Back in the day, if someone wanted to send mail overseas, they would need an international reply to coupon. Ponzi bought reply to coupons from different countries as they were cheaper and sold them in the States where they had a higher sell value. The difference was the profit he and his investors would share. He promised investors a 50% return in just 45 days, attracting thousands of local investors. At its height, he was gaining \$250,000 a day. The Ponzi scheme works when a con artist promises high returns with no risk to investors. However, the cost and time of conducting international business was adding up, so he could not pay investors. He decided to pay early investors they

While he was not the first financial criminal, the impact of the Ponzi scheme
The impact of the Ponzi scheme was crazy. half a dozen banks crashed after

Commented [SK1]: Add image of ponzi

Commented [SK2R1]: [How Ponzi Schemes Work | HowStuffWorks](#)
[The Rise and Fall of Charles Ponzi: How a Pyramid Scheme Changed the Face of Finance Forever | History Hit](#)

The Digital Era

As the internet and technology evolved, scams started using old tricks on new media. Within these Eras, we can find microtrends for scams, can be found))))))))))))))))))

[The Decade of Deception: How Scams Evolved from Basic Tricks to High-Tech Threats | by Sahana.hanji | Medium](#)

- 2013 - 2016 - Early digital scams

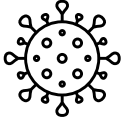
One of the earliest and most notable scams is the Nigerian prince scam, where victims would receive an email from a Nigerian prince requesting financial aid. In return for helping them out, victims would receive large compensation. However, once the victims transferred money, the “prince” would disappear. These scams worked because people were unaware that this was a scam, as the internet was still a relatively safe and trustworthy place. The scam was simple and effective because it emotionally manipulated victims, relying on their lack of awareness.

Dear Sir,
I am prince [redacted] from Nigeria. Your help would be very appreciated.
I want to transfer all of my fortune outside if Nigeria due to a frozen account,
If you could be so kind and transfer small sum of 3 500 USD to my account,
I would be able to unfreeze my account and transfer my money outside of
Nigeria. To repay your kindness, I will send 1 000 000 USD to your account.

Please contact me to proceed

Prince [redacted]

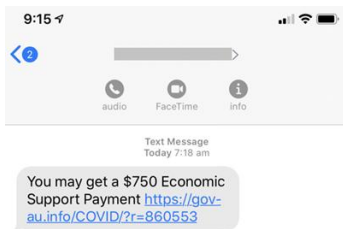
- 2017 - The rise of cryptocurrency
- With the rise of cryptocurrency, it is only natural there is a rise in cryptocurrency scams. OneCoin was a cryptocurrency (2014-2017) that defrauded investors of \$4 billion. What made people convinced of its legitimacy was its use of promotional material, fake endorsements from celebrities and fake events to lure investors in. The appeal of “being the next Bitcoin” and FOMO, made them ignore warning signs.



- 2020 - The Pandemic

The pandemic was an unprecedented time worsen scam. As the world suddenly shifted online, scammers adapted to the new environment. Scammers played on peoples fear and uncertainty, offering fake PPE, vaccines, and posing as government agencies offering financial aid.

The following examples were posted on ScamWatch, a website run by the ACCC, provides information to help people recognise, avoid and report scams.



- 2020-2022 - AI scams

With AI being incorporated into phishing scams, it's becoming increasingly harder to differentiate legitimate emails from fraudulent ones. As AI is able to be trained using a data sample from emails, it is able to make emails that use the same language, tone and specific details that we would use to distinguish an authentic email. In 2020, Google reported blocking over 18 million COVID-19 phishing emails per day, many of which used AI.



The Future of Scams

As technology continues to evolve, scammers have the ability to use it to their benefit. Not only will scams become more deceptive, but it will make it easier for scammers to hide from law enforcement. AI (Artificial Intelligence) and ML (Machine Learning) have started to be incorporated into scams with uses such as deepfake images, AI-generated phishing emails and texts.

- Deepfake ai-generated

Commented [SK3]: The Totally Fascinating History of Fraud - MoneyThumb
The Evolution of Modern Scams: Trends and Prevention Tips | Scam Prevention | Scam.SG

- Qr code scams
- blockchain

As digital space begin to grow, with companies like Meta creating the Metaverse with hyper realistic avatars, we can only expect that scams will reach those technologies as well.

TYPES OF SCAMS (1000 WORDS)

Scams are everywhere, and I mean that literally. They are embedded in our current society, a 7-year-old would be able to identify a common email scam. BUT, while email scams might be one of the most common, the sneakier and more dangerous scams get overshadowed.

The Fishy Phishing Scams

- Email Scams
- SMS Scams
- Fake Login in Pages



The Deadly Investment Scam

Investment scams are scams where the attacker claims you make big returns if you invest x amount of money into their business. The business that you'll likely invest in financial markets, cryptocurrency, real estate and precious metals. Scammers get your attention through social media posts, online ads, free events or a free course that promises to teach you the secret to getting rich. Australians have lost over \$1.3 billion in investments scams - scamwatch.



Investment scams can include Ponzi Schemes

Scam Study: Spanish Prisoner Scam

The Spanish prisoner trick is a scam originating in the early 19th century. It is when the scammer tells his victim that he is a wealthy person who has been imprisoned in Spain under a false identity. In exchange for money, the scammer not only offers a greater monetary reward, but the chance to marry his beautiful daughter. Once the victim has given the scammer, suddenly more money is needed



The Nefarious Romance Scam

The romance scam is when a scammer uses a fake online identity to gain the victims trust and affection in the hopes, they provide money or personal information. Romance scams usually occur on social media platforms, emails and SMS. While we might think romance scams developed with the introduction of technology within

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Hanna-S-Cunha-Article.pdf

the dating space; the first romance scam can be dated back to the 16th century, known as the “Spanish Prisoner” scam.

This type of scam is so effective because the victims believe that the scammers have strong feeling for them, rather than the desire for their money.

The Tricky Tech Scam

- **Social media scam**

Social media scams are scams that exploit users on social media into giving personal information or money. The difference from other phishing attacks is that these scams happen on social media platforms. Common types of social media scams are:



Commented [SK5]: [Common social media scams to avoid](#)

- Impersonation scams, where an attacker creates a fake profile pretending to be someone you know to gain your trust
- Fake sales, where scammers promote a fake product or service to trick users into making a purchase
- Quizzes and competitions, where scammers create “fun” quizzes to collect users’ personal information.

Romance scams and phishing scams can also occur on social media, making it imperative to know who you are talking to on social media, and monitor what information you are sharing with others.

- **AI Generated Deep Fakes**

As technology continues to embed itself into our society and is constantly improving and evolving, we need to remain educated and aware of how technology can be abused by scammers. Deep fakes are images, videos or audio that has been edited or generated with the use of artificial intelligence. Hyper realistic fake videos and audios are becoming more common, with deepfake attacks happening ever 5 minutes in 2024, making up 40% of biometric fraud.

Commented [SK6]: [Deepfake Scams Are Stealing Millions—How To Spot One](#)

However, despite these statistics, scammers are able to exploit victims because deepfakes play on our trusting nature. When we see a familiar face, or hear a recognisable voice, we do not assume it is a scam. This coupled with a sense of urgency and secrecy created by the attacker, “*I need money right now, I am in trouble, you can’t tell*

anyone", we get emotionally manipulated into making a quick decision that bypasses our usual scepticism.

In 2023, an employee in Hong Kong joined a video conference with his company's CFO, which resulted in him transferring \$25 million to a fraudulent bank account. But if his boss told him to, what's the issue? It was not the CFO on call - scammers had used AI to create deepfake video of the executive, deceiving the employee and manipulating him into transferring the funds.

Scam Study: WannaCry Ransomware

WannaCry is a ransomware worm (a worm is malicious software that automatically spreads itself to computers on a network) that spread through computers in May 2017, impacting more than 200,000 computers in over 150 countries. It infected Windows computers, encrypted all the files on the hard drive, and demanded a ransom payment in bitcoin in order to decrypt the files. What made WannaCry dangerous is the number of important systems impacted, such as the British National Health Service, FedEx, Honda, Nissan. The virus spread using a vulnerability exploit "Eternal Blue", which was developed by the NSA, but stolen by a group called the Shadow Brokers. Within hours, the

The Ransomware scam

Ransomware scam is when malicious software locks or encrypts your data so you can no longer access them. To recover your data, a ransom, usually in the form of cryptocurrency is demanded.

While ransomware scams are more common for organisations, individuals can still fall victim to them.

If you do not have a backup of your data, it may be nearly impossible to recover your data. Ransomware scams are a tricky situation since you can never be sure that paying the ransom will allow you to regain access to your data, and that the data has not already been sold or leaked online.

The Australian Signals Directorate (ASD), a national organisation, advises to never pay a ransom in the case of a ransomware scam.

The In Person Scam

In a world full of technology, it's easy to forget that scams can also occur in person. Scammers may knock on your door or approach you in public and:

- Ask for upfront payment for a goods and services
- Pressure you in completing an online survey to get your personal details
- Donations to a "charity"

Statistics

WHY WE ARE SUSCEPTIBLE TO SCAMS – (700 WORDS)

Humans are the weakest link.

- Scammers understand this fact, and manipulate
- Emotional manipulation
- Cognitive biases
- Social influence
- Lack of awareness

As scamming techniques keep evolving, scammers are able to play on victim's lack of knowledge in a certain topic such as cryptocurrencies, blockchain,

ANATOMY OF A SCAMS – HOW SCAMMERS GET TO YOU – (800 WORDS)

Phishing Scams

All types of phishing scams, whether it be SMS or email scams work the same way, the only difference is the delivery.

For phishing attacks, attackers can control 2 variables for the type of attack they are going to launch.

1. The attacker must decide what they want to achieve from this scam; steal credit card information, steal passwords,..... For example, the attacker wants to get credit card information from the employees of "Radius".
2. The attacker must then choose who they are going to target. Even for the simplest phishing emails, the attacker needs to know who they are going to send it to, For example a specific company, the contacts of someone already compromised, emails that they have been they have bought.

From a previous attack, the scammer already has the personal emails of all the employees of the company.

3. Do a background check. Once the target has been selected, the scammer needs to figure out how they are going to launch their attack. They need to select the medium of the attack, for example, SMS, email, social media, and the type of phishing attack, for example, spear phishing, mass targeting attack.

Since the attacker wants as many credit card information as possible, they decided to launch an email mass target phishing email

4. Launch the attack. Regardless of the attack type, the attacker will always use a compromised system to distribute the phishing email. For example, someone may have been previously a victim to a phishing attack.
5. Hook the victims. This is where the attacker will manipulate the victim into clicking to an embedded link. They think they are clicking to
In the email, there will be a call to action stating that the user can "claim their free gift"
6. Expand and monetise. Once the user has clicked on a link, the attacker is able to get the victim's username and password, through a convincing login page. With this information they are able to gain access to data, launch more attacks on the victim's contacts _____.

And they now have credit card information. WOOOOOOOOOOOOOOOOOOO

Romance Scams

This scam usually lasts about 6-8 months.

1. The first phase is when a fake profile is created through social media or an online dating site or app. They often use low quality images that are heavily pixelated.
2. The second phase is contacting the target. The scammer initiates contact with the victim with the purpose of creating a strong bond filled with trust, confidence and romantic connection.
3. The third phase is asking for money. Suddenly a tragedy, or theft of personal documents, or unexpected hospital fees or travel costs to meet the victim will cause the scammer to seek money from the victim.
4. Repeat phase 3. The scammer will continue to be overcome with personal tragedy until the victim realises, they are being misled and stops sending them money.

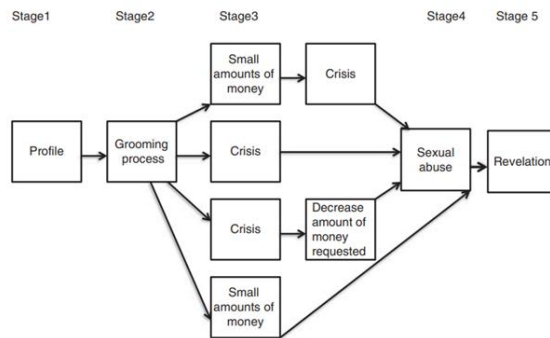


Figure 1: Romance scam trajectories.

Fake Payment Scam

1. Scammers get you name and email address, and physical address or phone number
2. They then write up a scam email or letter
3. You receive the email or letter or call
4. You then contact the scammer and give them your personal information
5. They send counterfeit checks
 - a. NEVER CASH A CHECK

EFFECT OF SCAMS ON VICTIMS AND SCAM TARGET ARCHETYPES

Scam Target Archetypes

-----Why certain groups get targeted and how scammers tailor their approach.-----

Scammers usually target the weak, such as older people who don't know how to use technology, young people who are looking to make a quick dollar



EFFECT OF SCAMS ON THE ENVIRONMENT

When we think of scams, we think of fraudulent phishing emails or SMS messages, however, we rarely consider their environmental impact.

Phishing websites are hosted on servers, which are located at data centres. Data centres are essentially a large warehouse where thousands of servers are stored, which need consistent access to electricity, air conditioning and cooling systems. They consume 1-1.5% of the world's global electricity per year and generate 1% of energy related greenhouse gas emissions. Since the introduction of AI technologies, the energy demand on AI data centres is rapidly increasing, with these data centres using 5-10 times more energy than traditional data centres.



What does this mean for the environment?

Well to begin, data centres require large amounts of land. Data centres are usually built in rural regions, resulting in deforestation or disruption of the local environment. If they are built in urban locations, due to the compactness, the urban data centres contribute to the urban heat island effect, due to the concentrated heat emissions.

Data centres need to constantly upgrade their equipment to stay competitive, which leads to high amounts of electronic waste (E-waste) being produced. Technologies such as servers, cooling units, power supplies and cables need to be replaced every 2-5 years.

How It Relates to Scams

HOW TO STAY PROTECTED – (900 WORDS)

As a response to the uptick in scams, government agencies have started creating regulatory bodies, such as the ACCC's ScamWatch in Australia or the European Union's General Data Protection Regulation, making it more difficult for scammers to access and misuse data.

In cyber security, humans will always be the weakest link. At the point in cyber security history, it's a fact we need to be cautious about. Thus, the only solution is a human one. As scam detection technology is constantly evolving, so are scams. We are the answer to staying safe from scams,

To stay one step ahead of scammers just remember -

A.U.S.G.U.T.

AUSGUT is a list of questions we should ask ourselves if we find an email, SMS, phone call suspicious.

Is the scammer A: Asking for personal information?

Does the message have a dodgy U: URL?

Does the scam message elicit S: sense of urgency?

Does the message have a strange G: Greeting?

Does the message ask for an U: an unusual payment request?

Is T: The email address suspicious looking?

As scam technology evolves, we need to start protecting and educating ourselves from AI scams. While AUSGUT principles still apply to AI scams, there is an added element of deepfake technology being used. To protect yourself from AI scammers, you need to focus on what you see.

- Is the visuals in the video off? Is the lips not syncing to the words that are coming out of their mouths?
- Do they have an unusual voice pattern, are they speaking monotonously?
- Are they making excuses not to show their face?

SIMULATION – (600 WORDS)

TAKEAWAY